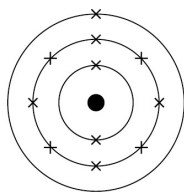


4.5.1 The periodic table

The model of atomic structure introduced in [Atomic structure](#) (page 24) is further developed and applied here. The arrangement of elements in the periodic table can be explained in terms of atomic structure, which is evidence for the model of a nuclear atom with electrons in energy levels. The periodic table organises the known chemical elements in a way that helps to account for their physical and chemical properties.

The focus is on the elements in groups 1, 7 and 0.

4.5.1.1 Atomic number and the periodic table

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the position of an element in the periodic table is related to the arrangement of electrons in its atoms and hence to its atomic number.	The elements in the periodic table are arranged in order of atomic (proton) number, so that elements with similar properties are in columns known as groups. The table is called a periodic table because similar properties occur at regular intervals. Electrons occupy particular energy levels. Each electron in an atom is at a particular energy level (in a particular shell). The electrons in an atom occupy the lowest available energy levels (innermost available shells). Elements in the same group in the periodic table have the same number of electrons in their outer shell (outer electrons) and this gives them similar chemical properties.	WS 1.2 Represent the electronic structure of the first 20 elements of the periodic table in the following forms:  sodium 2,8,1 Predict possible reactions and probable reactivity of elements from their positions in the periodic table. This topic links with Atomic structure (page 24).
Explain in terms of isotopes how this changes the arrangement proposed by Mendeleev.	Following Mendeleev, the elements in the periodic table were arranged in order of relative atomic mass. In this order some elements appeared to be in the wrong group. These problems were solved once it was realised that most elements occur as mixtures of isotopes and that elements should be arranged in the table in order of atomic number.	WS 1.1 Show how scientific methods and theories have changed over time. This topic links with Atomic structure (page 24).

4.5.1.2 Metals and non-metals

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the atomic structure of metals and non-metals relates to their position in the periodic table. Explain how the reactions of elements are related to the arrangement of electrons in their atoms and hence to their atomic number.	The majority of elements are metals. Metals are found to the left and towards the bottom of the periodic table. Non-metals are found towards the right and top of the periodic table. Elements that react by losing their outer electrons to form positive ions are metals. Elements that do not form positive ions are non-metals. The more reactive non-metals, such as the halogens, react with metals by gaining electrons to form negative ions.	WS 1.2 Describe metals and non-metals and explain the differences between them in terms of their characteristic physical and chemical properties (see Structure and bonding (page 98) and the sections about groups 1, 7 and 0 in this topic).

4.5.1.3 Group 0

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall the simple properties of Group 0. Explain how the observed simple properties of Group 0 depend on the outer shell of electrons of the atoms and predict properties from given trends down the group.	The elements in Group 0 of the periodic table are called the noble gases. They are unreactive and do not easily form molecules because their atoms have stable arrangements of electrons. The noble gases have eight electrons in their highest energy level (outer shell), except for helium, which has only two electrons. The boiling points of the noble gases increase with increasing relative atomic mass (going down the group).	WS 1.2 Predict properties from given trends down Group 0.

4.5.1.4 Group 1

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall the simple properties of Group 1. Explain how the observed simple properties of Group 1 depend on the outer shell of electrons of the atoms and predict properties from given trends down the group.	The elements in Group 1 of the periodic table are known as the alkali metals. They: • are soft metals with low density • react with non-metals, including chlorine and oxygen, to form colourless ionic compounds • react with water • form hydroxides that give alkaline solutions in water. In Group 1, the further down the group an element is, the more reactive the element.	WS 1.2 Predict properties from given trends down Group 1.

4.5.1.5 Group 7

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall the simple properties of Group 7. Explain how the observed simple properties of Group 7 depend on the outer shell of electrons of the atoms and predict properties from given trends down the group.	The elements in Group 7 of the periodic table are known as the halogens. They: • are non-metals • consist of molecules • react with metals to form ionic compounds • form molecular compounds with other non-metallic elements. In Group 7, the further down the group an element is the higher its relative molecular mass, melting point and boiling point. In Group 7, reactivity of the elements decreases going down the group. A more reactive halogen can displace a less reactive halogen from an aqueous solution of its salt.	WS 1.2 Predict properties from given trends down Group 7.